Category
Analysis of Inventories

Related Questions:
Practice questions related to this topic

INVENTORY VALUATION

Learning Outcome

- describe the measurement of inventory at the lower of cost and net realisable value and its implications for financial statements and ratios

Significant financial risk can result from the holding of inventory. The cost of inventory may not be recoverable due to spoilage, obsolescence, or declines in selling prices. IFRS states that inventories shall be measured (and carried on the balance sheet) at the lower of cost and net realizable value.[1] Net realizable value is the estimated selling price in the ordinary course of business, less the estimated costs necessary to make the sale and estimated costs to get the inventory in condition for sale. The assessment of net realizable value is typically done item by item or by groups of similar or related items. In the event that the value of inventory declines below the carrying amount on the balance sheet, the inventory carrying amount must be written down to its net realizable value[2] and the loss (reduction in value) recognized as an expense on the income statement. This expense may be included as part of cost of sales or reported separately.

In each subsequent period, a new assessment of net realizable value is made. Reversal (limited to the amount of the original write-down) is required for a subsequent increase in value of inventory previously written down. The reversal of any write-down of inventories is recognized as a reduction in cost of sales (reduction in the amount of inventories recognized as an expense).

US GAAP used to specify the lower of cost or market to value inventories.[3] For fiscal years beginning after 15 December 2016, inventories measured using other than last-in, first-out (LIFO) and retail inventory methods are measured at the lower of cost or net realizable value. This is broadly consistent with IFRS with one major difference: US GAAP prohibits the reversal of write-downs. For inventories measured using LIFO and retail inventory methods, market value is defined as current replacement cost subject to upper and lower limits. Market value cannot exceed net realizable value (i.e., the selling price less reasonably estimated costs of completion and disposal). The lower limit of market value is net realizable value less a normal profit margin. Any write-down to market value or net realizable value reduces the value of the inventory, and the loss in value (expense) generally is reflected in the income statement in the cost of goods sold.

An inventory write-down reduces both profit and the carrying amount of inventory on the balance sheet and thus has a negative effect on profitability, liquidity, and solvency ratios. However, activity ratios (e.g., inventory turnover and total asset turnover) will be positively affected by a write-down because the asset base (denominator) is reduced. The negative impact on some key ratios, due to the decrease in profit, may result in the reluctance by some companies to record inventory write-downs unless evidence is strong that the decline in the value of inventory is permanent. This is especially true under US GAAP, in which case reversal of a write-down is prohibited.

International Accounting Standards 2 (IAS 2), *Inventories*, does not apply to the inventories of producers of agricultural and forest products and minerals and mineral products, nor to commodity broker–traders. These inventories may be measured at net realizable value (fair value less costs to sell and complete) according to well-established industry practices. If an active market exists for these products, the quoted market price in that market is the appropriate basis for determining the fair value of that asset. If an active market does not exist, a company may use market determined prices or values (such as the most recent market transaction price) when available for determining fair value. Changes in the value of inventory (increase or decrease) are recognized in profit or loss in

EXAMPLE 1

Rate Your Confidence

Accounting for Declines and Recoveries of Inventory Value

Hatsumei Enterprises, a hypothetical company, manufactures computers and prepares its financial statements in accordance with IFRS. In 2017, the cost of ending inventory was EUR5.2 million, but its net realizable value was EUR4.9 million. The current replacement cost of the inventory is EUR4.7 million. This figure exceeds the net realizable value less a normal profit margin. In 2018, the net realizable value of Hatsumei's inventory was EUR0.5 million greater than the carrying amount.

Category
Analysis of Inventories

Related Questions:
Practice questions related to this topic

1. What was the effect of the write-down on Hatsumei's 2017 financial statements? What was the effect of the recovery on Hatsumei's 2018 financial statements?

   Hide Solution

   Solution:

   For 2017, Hatsumei would write down its inventory to EUR4.9 million and record the change in value of EUR0.3 million as an expense on the income statement. For 2018, Hatsumei would increase the carrying amount of its inventory and reduce the cost of sales by EUR0.3 million (the recovery is limited to the amount of the original write-down).

2. Under US GAAP, if Hatsumei used the LIFO method, what would be the effects of the write-down on Hatsumei's 2017 financial statements and of the recovery on Hatsumei's 2018 financial statements?

   Hide Solution

   Solution:

   Under US GAAP, for 2017, Hatsumei would write down its inventory to EUR4.7 million and typically include the change in value of EUR0.5 million in cost of goods sold on the income statement. For 2018, Hatsumei would not reverse the write-down.

3. What would be the effect of the recovery on Hatsumei's 2018 financial statements if Hatsumei's inventory were agricultural products instead of computers?

   Hide Solution

   Solution:

   If Hatsumei's inventory were agricultural products instead of computers, inventory would be measured at net realizable value and Hatsumei, therefore, would increase inventory and record a gain of EUR0.5 million for 2018.

Analysts should consider the possibility of an inventory write-down because the impact on a company's financial ratios may be substantial. The potential for inventory write-downs can be high for companies in industries in which technological obsolescence of inventories is a significant risk. Analysts should carefully evaluate prospective inventory impairments (as well as other potential asset impairments) and their potential effects on the financial ratios when debt covenants include financial ratio requirements. The breaching of debt covenants can have a significant impact on a company.

Companies that use specific identification, weighted average cost, or FIFO methods are more likely to incur inventory write-downs than companies that use the LIFO method.

Thus, it is far less likely that inventory write-downs will occur under LIFO—and if a write-down does occur, it is likely to be of a lesser magnitude.

EXAMPLE 2

Effect of Inventory Write-Downs on Financial Ratios

The Volvo Group, based in Göteborg, Sweden, is a leading supplier of commercial transport products, such as construction equipment, trucks, busses, and drive systems for marine and industrial applications as well as aircraft engine components.[4] Excerpts from Volvo's consolidated financial statements are shown in Exhibits 1 and 2. Notes pertaining to Volvo's inventories are presented in Exhibit 3.

Exhibit 1:

Volvo Group Consolidated Income Statements (Swedish krona in millions, except per share data)

For the years ended 31 December	2017	2016	2015
Net sales	334,748	301,914	312,515
Cost of sales	(254,581)	(231,602)	(240,653)
Gross income	80,167	70,312	71,862
⋮	⋮	⋮	⋮
Operating income	30,327	20,826	23,318
Interest income and similar credits	164	240	257
Income expenses and similar charges	(1,852)	(1,847)	(2,366)
Other financial income and expenses	(386)	11	(792)
Income after financial items	28,254	19,230	20,418
Income taxes	(6,971)	(6,008)	(5,320)
Income for the period	21,283	13,223	15,099
Attributable to:			
Equity holders of the parent company	20,981	13,147	15,058
Minority interests	302	76	41
Profit	21,283	13,223	15,099

Exhibit 2:

Volvo Group Consolidated Balance Sheets (Swedish krona in millions)

31 December	2017	2016	2015
Assets			
Total non-current assets	213,455	218,465	203,478
Current assets:			
Inventories	52,701	48,287	44,390
⋮	⋮	⋮	⋮
Cash and cash equivalents	36,092	23,949	21,048
Total current assets	199,039	180,301	170,687
Total assets	412,494	398,916	374,165
Shareholders' equity and liabilities			

Minority interests	1,941	1,703	1,801
Total shareholders' equity	109,011	97,764	85,610
Total non-current provisions	29,147	29,744	26,704
Total non-current liabilities	96,213	104,873	91,814
Total current provisions	10,806	11,333	14,176
Total current liabilities	167,317	155,202	155,860
Total shareholders' equity and liabilities	412,404	398,916	374,165

Exhibit 3:

Volvo Group Selected Notes to Consolidated Financial Statements

Note 17. Inventories

Accounting Policy

Inventories are reported at the lower of cost and net realizable value. The cost is established using the first-in, first-out principle (FIFO) and is based on the standard cost method, including costs for all direct manufacturing expenses and the attributable share of capacity and other related manufacturing-related costs. The standard costs are tested regularly and adjustments are made based on current conditions. Costs for research and development, selling, administration and financial expenses are not included. Net realizable value is calculated as the selling price less costs attributable to the sale.

Sources of Estimation Uncertainty

Inventory obsolescence

If the net realizable value is lower than cost, a valuation allowance is established for inventory obsolescence. The total inventory value, net of inventory obsolescence allowance, was SEK52,701 (in millions) as of December 2017, and SEK48,287 as of 31 December 2016.

Panel A: Inventory

31 December (millions of krona)	2017	2016	2015
Finished products	32,304	31,012	27,496
Production materials, etc.	20,397	17,275	16,894
Total	**52,701**	**48,287**	**44,390**

Panel B: Increase (decrease) in allowance for inventory obsolescence

31 December (millions of krona)	2017	2016	2015
Opening balance	3,683	3,624	3,394
Change in allowance for inventory obsolescence charged to income	304	480	675
Scrapping	(391)	(576)	(435)
Translation differences	(116)	177	(29)
Reclassifications, etc.	8	(23)	20

1. What inventory values would Volvo have reported for 2017, 2016, and 2015 if it had no allowance for inventory obsolescence?

Hide Solution

Solution:

31 December (Swedish krona in millions)	2017	2016	2015
Total inventories, net	52,701	48,287	44,390
From Note 17. (Allowance for obsolescence)	3,489	3,683	3,624
Total inventories (without allowance)	56,190	51,970	48,014

2. Assuming that any changes to the allowance for inventory obsolescence are reflected in the cost of sales, what amount would Volvo's cost of sales be for 2017 and 2016 if it had not recorded inventory write-downs in 2017 and 2016?

Hide Solution

Solution:

31 December (Swedish krona in millions)	2017	2016
Cost of sales	254,581	231,602
(Increase) decrease in allowance for obsolescence*	194	(59)
Cost of sales without allowance	254,775	231,543

* From Note 17, the decrease in the allowance for obsolescence for 2017 is 194 (3,489 – 3,683) and the increase for 2016 is 59 (3,683 – 3,624).

3. What amount would Volvo's profit (net income) be for 2017 and 2016 if it had not recorded inventory write-downs in 2017 and 2016? Volvo's effective income tax rate was reported as 25 percent for 2017 and 31 percent for 2016.

Hide Solution

Solution:

31 December (Swedish krona in millions)	2017	2016
Profit (Net income)	21,283	13,223
Increase (reduction) in cost of sales	(194)	59
Taxes (tax reduction) on operating profit*	49	(18)
Profit (without allowance)	21,138	13,264

* Taxes (tax deductions) on the operating profit are assumed to be 49 (194 x 25%) for 2017 and –18 (–59 x 31%) for 2016.

4. What would Volvo's 2017 profit (net income) have been if it had reversed all past inventory write-downs in 2017? This question is independent of 1, 2, and 3. The effective income tax rate was 25 percent for 2017.

Hide Solution

Solution:

31 December (Swedish krona in millions)	2017
Profit (Net income)	21,283
Reduction in cost of sales (increase in operating profit)	3,489
Taxes on increased operating profit*	−872
Profit (after recovery of previous write-downs)	23,900

Rate Your Confidence

Category
Analysis of Inventories

Related Questions:
Practice questions related to this topic

assuming no allowance for inventory obsolescence as in questions 1, 2, and 3: inventory turnover ratio, days of inventory on hand, gross profit margin, and net profit margin.

Hide Solution

Solution:

The Volvo Group's financial ratios for 2017 with the allowance for inventory obsolescence and without the allowance for inventory obsolescence are as follows:

	With Allowance (As Reported)	Without Allowance (Adjusted)
Inventory turnover ratio	5.04	4.71
Days of inventory on hand	72.4	77.5
Gross profit margin	23.95%	23.89%
Net profit margin	6.36%	6.31%

Inventory turnover ratio = Cost of sales ÷ Average inventory

With allowance (as reported) = 5.04 = 254,581 ÷ [(52,701 + 48,287) ÷ 2]

Without allowance (adjusted) = 4.71 = 254,775 ÷ [(56,190 + 51,970) ÷ 2]

Inventory turnover is higher based on the numbers as reported because inventory carrying amounts will be lower with an allowance for inventory obsolescence. The company might appear to manage its inventory more efficiently when it has inventory write-downs.

Days of inventory on hand = Number of days in period ÷ Inventory turnover ratio

With allowance (as reported) = 72.4 days = (365 days ÷ 5.04)

Without allowance (adjusted) = 77.5 days = (365 days ÷ 4.71)

Days of inventory on hand are lower based on the numbers as reported because the inventory turnover is higher. A company with inventory write-downs might appear to manage its inventory more effectively. This is primarily the result of the lower inventory carrying amounts.

Gross profit margin = Gross profit ÷ Net sales

With allowance (as reported) = 23.95% = (80,167 ÷ 334,748)

Without allowance (adjusted) = 23.89% = [(80,167 − 194) ÷ 334,748]

In this instance, the gross profit margin is slightly higher with inventory write-downs because the cost of sales is lower (due to the reduction in the allowance for inventory obsolescence). This assumes that inventory write-downs (and inventory write-down recoveries) are reported as part of cost of sales.

Net profit margin = Profit ÷ Net sales

With allowance (as reported) = 6.36% = (21,283 ÷ 334,748)

Without allowance (adjusted) = 6.31% = (21,138 ÷ 334,748)

In this instance, the net profit margin is higher with inventory write-downs because the cost of sales is lower (due to the reduction in the allowance for inventory obsolescence). The absolute percentage difference is less than that of the gross profit margin because of the income tax reduction on the decreased income without write-downs.

would appear less attractive without the write-downs. The inventory turnover ratio is slightly better (higher) with inventory write-downs because inventory write-downs decrease the average inventory (denominator), making inventory management appear more efficient with write-downs.

Rate Your Confidence

Discuss Filter

Discussion

What do you want to discuss... (required)

RC For questions that have the potential of write-down reversal but no specification for IFRS or US GAAP, are we to just assume it's IFRS? I've gotten several of these wrong by assuming US GAAP

Created 6 months ago by Ran Cheng 1 reply | Last Activity: 3 months ago

Show All replies Reply to this Comment

MP why in question 3 we take net income instead of plugging for example for year 2017 into cost of sales the increase of 194? Tax isbefore net income so that is why we cannot substract from net income and calculate another tax

Created 7 months ago by Manuel Paredes Caldito 3 replies | Last Activity: 7 months ago

Show All replies Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic

Rate Your Confidence

Category

Analysis of Inventories

Related Questions:

Practice questions related to this topic